Interpretability and Missing Data

MFE5340: AI in FE: Quantitative Investment

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Introduction

1 Why Feature Importance

- Machine learning (ML) models are often seen as “black boxes.”
- In asset pricing, interpretability is critical for:
  - Understanding drivers of stock returns.
  - Building trust with stakeholders.
  - Ensuring regulatory compliance.
- Feature importance helps us identify the most influential factors in ML models.
- A great book on this topic: Christoph Molnar, *Interpretable Machine Learning*, https://christophm.github.io/interpretable-ml-book/

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Types of Feature Importance

	Model-Specific Methods	Model-Agnostic Methods
Definition	Designed for specific models (e.g., linear regression, decision trees).	Applicable to any machine learning model, regardless of its type.
Flexibility	Limited to the specific model it is designed for.	Highly flexible and can be used with any model.
Examples	Coefficients in linear regression, feature importance in decision trees.	Permutation feature importance, partial dependence plots, Shapley values.
Computational Cost	Usually low, as it leverages the model's structure.	Can be computationally expensive, especially for large datasets or complex models.
Use Cases	Best for simple, interpretable models.	Ideal for complex or black-box models.

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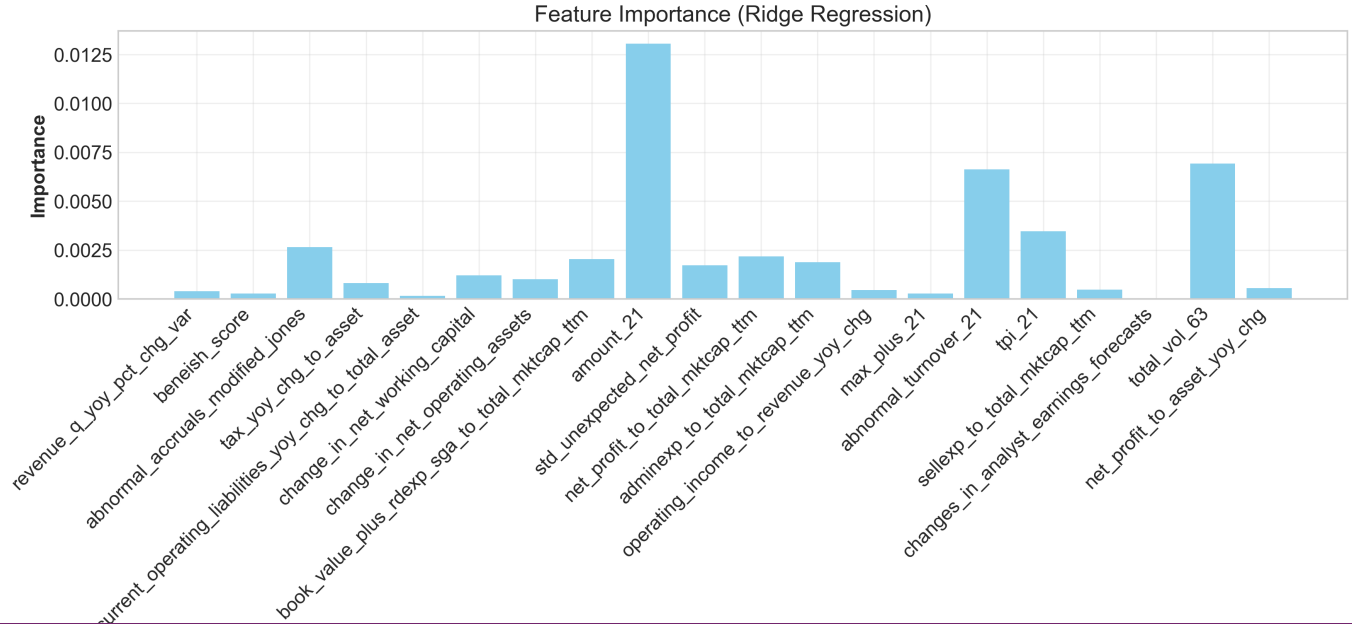
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- Linear models include OLS, ridge regression, lasso regression, etc. (Recall the linear model lecture!)
- When features are standardized, i.e., they are on the same scale, then the importance of feature k is $|\beta_k|$.
- When features are no standardized, i.e., they may not be on the same scale, then the importance of feature k is $\sigma(x_k)|\beta_k|$.
- **Warning!** For OLS, this is problematic with high multicollinearity. (Recall the ROE, ROA, ROIC example!)
- Example: 20 features, 93,746 stock-month samples. Let's apply ridge regression (λ set to 5.0 for illustration).

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Example – Ridge Regression

2 Model-Specific Methods

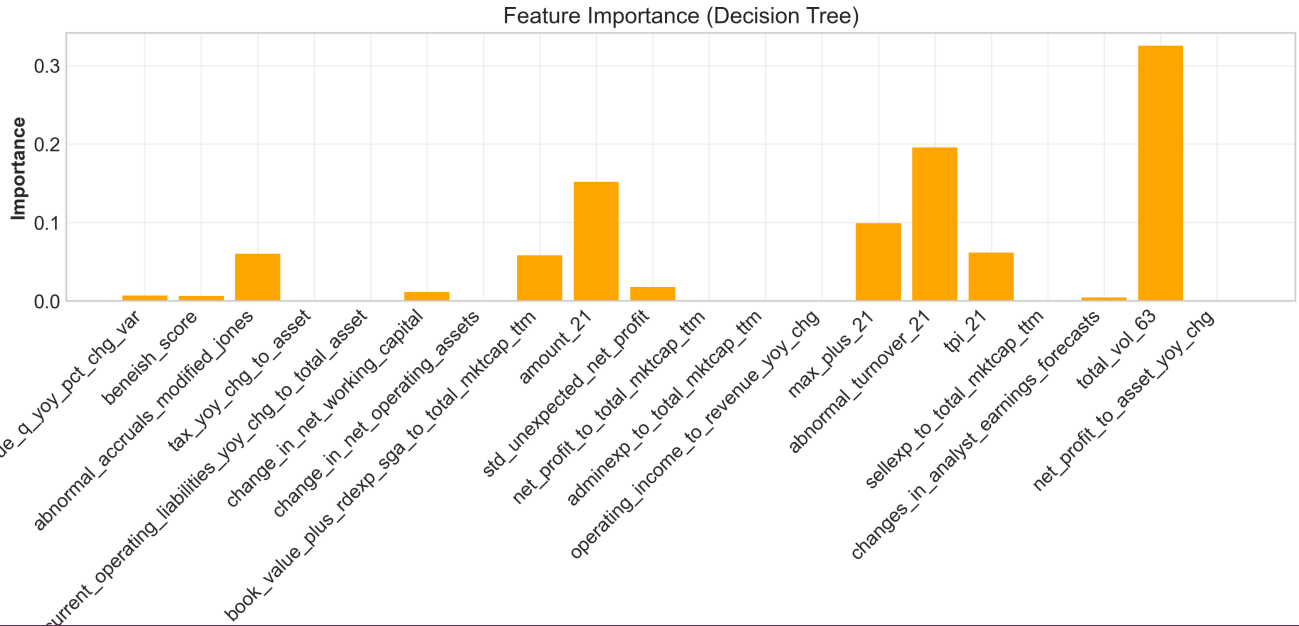


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- Tree-based models include regression tree, random forest, and boosting. (Recall the nonlinear model lecture!)
- Simple measures:
  - Number of times a feature is selected as the splitting variable.
  - Reduction in impurity (measured by MSE for example) when splitting on variable.
- Sophisticated measures:
  - Increase in prediction error on "out-of-bag" samples after permuting feature.
- Same example...

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Example – Decision Tree



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Global vs. Local

	Global Methods	Local Methods
Scope	Explains the entire model behavior across all data points.	Explains the model's prediction for a single data point or a small subset of data.
Goal	Provide a holistic understanding of how the model works.	Understand why the model made a specific prediction for an individual instance.
Examples	Permutation Feature Importance, Partial Dependence Plot, Global Surrogate Models.	Local Surrogate Models (LIME), Shapley Values, SHAP (for single instance).
Advantages	Useful for debugging the model, understanding overall trends, and identifying global biases.	Provides detailed insights into specific predictions, which is useful for decision-making and fairness analysis.
Disadvantages	May miss nuances of individual predictions; less useful for understanding specific cases.	Limited to specific instances; may not generalize well to the entire model.

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Permutation Feature Importance: Intuition

- A global model-agnostic method to measure the importance of a feature.
- If a feature is important, **shuffling** its values will disrupt the model's predictions, leading to an increase in error. (If shuffling a feature does not change the model's performance, the feature is likely unimportant.)
- Steps:
  1. Measure the model's baseline error on the test set.
  2. Permute (shuffle) the values of one feature while keeping others unchanged.
  3. Measure the new error after shuffling.
  4. The difference between the two errors quantifies the feature's importance.

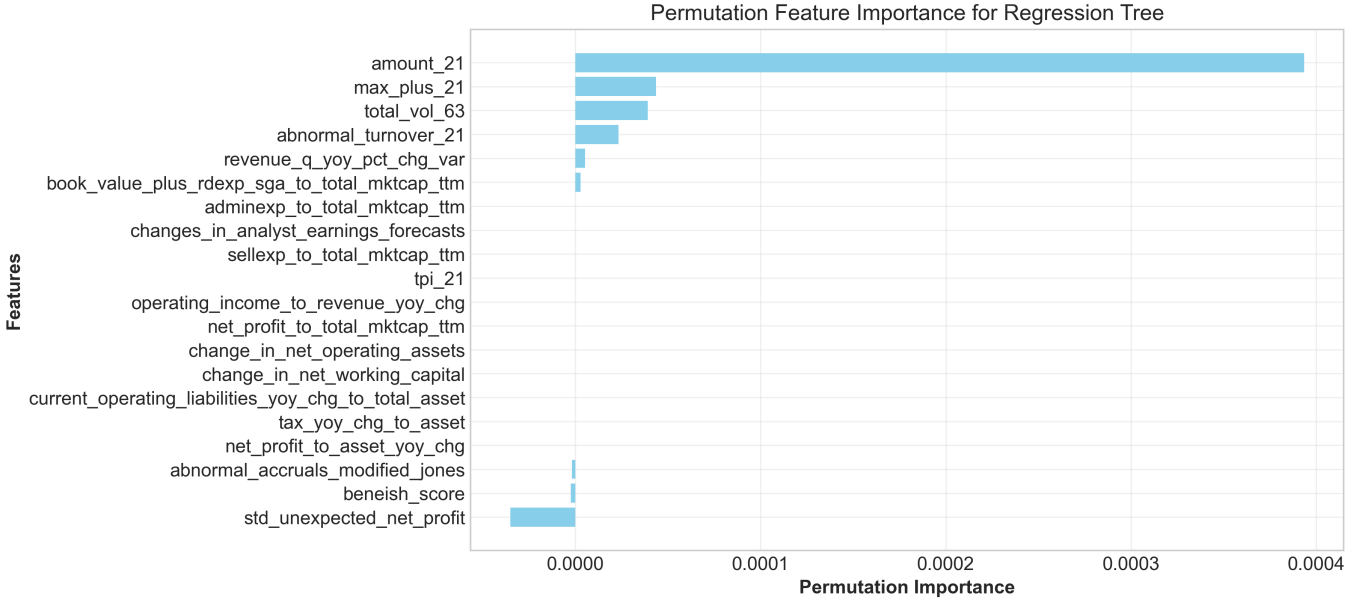
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Permutation Feature Importance: Math

- Let $l_{\text{orig}} = \mathcal{L}(y, \hat{f}(\boldsymbol{x}))$ be the baseline model error on the test set.
- For each feature k:
  1. Generate $\boldsymbol{x}_{\text{perm},k}$ by randomly permuting the values of feature k.
  2. Compute the error after shuffling:
  $$l_{\text{perm},k} = \mathcal{L}(y, \hat{f}(\boldsymbol{x}_{\text{perm},k})).$$
  3. Feature importance is defined as:
  $$\text{FI}_k = l_{\text{perm},k} - l_{\text{orig}}.$$
- Larger FI_k suggests that feature k has a greater impact on the model's predictions.

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Permutation Feature Importance: Example



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Partial Dependence Plot: Intuition

- A global model-agnostic method to visualize the relationship between features and predictions.
- PDP shows how the model's predictions change with respect to a specific feature x_k by calculating the average prediction across all possible values of other features.
- Steps:
  1. Select a target feature x_k.
  2. For each value of x_k, average the model's predictions over all possible values of other features.
  3. Plot the average predictions as a function of x_k.
- PDP helps interpret the marginal effect of features on the model's output. It is typically computed on the training set, as it reflects how the model has learned to use the features during training.

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Partial Dependence Plot: Math

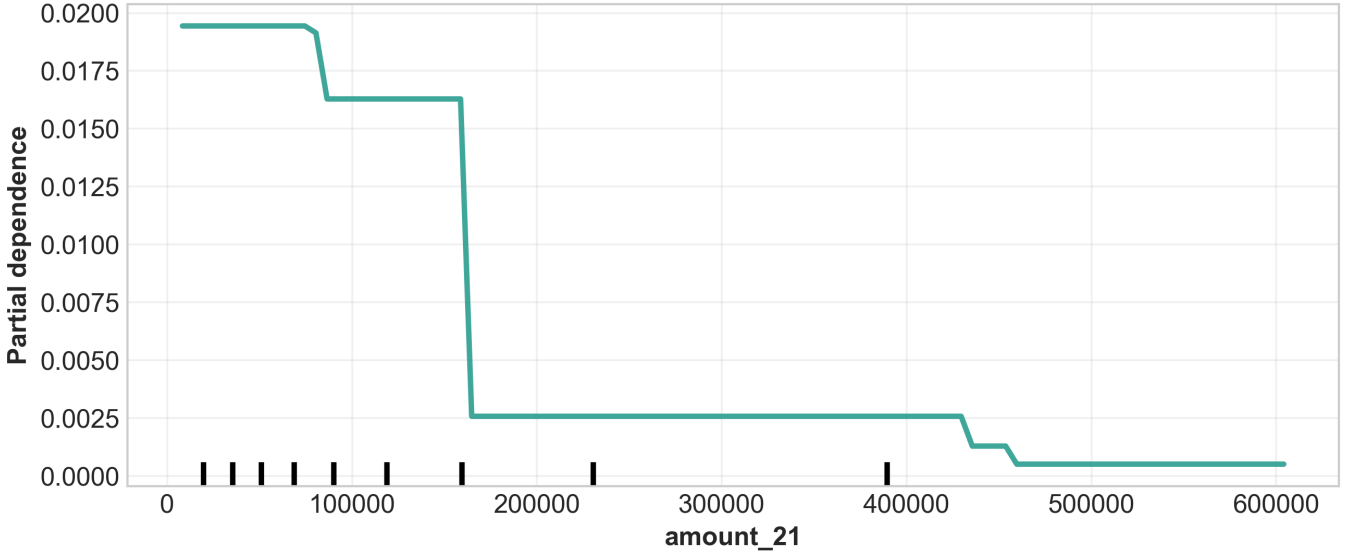
- Let $f(\boldsymbol{x})$ be the model's prediction function, and $\boldsymbol{x}_C$ denote all features except x_k.
- For feature x_k:

$$f_{\text{PDP}}(x_k) = \frac{1}{n} \sum_{i=1}^{n} f(x_k, \boldsymbol{x}_{C_i}),$$

  where $\boldsymbol{x}_{C_i}$ are the observed values of other features.
- PDP assumes independence between x_k and $\boldsymbol{x}_C$, which may not hold in correlated data.

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Partial Dependence Plot: Example



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Global Surrogate Models: Intuition

- A global model-agnostic method to approximate and interpret complex models by fitting a simpler, interpretable model (e.g., linear regression or decision tree). **The surrogate model is trained to approximate the behavior of the complex model by using the complex model's predictions as the target, rather than the original labels.**

- Steps:

  1. Select a complex model f as the target for approximation.
  2. Generate predictions $\hat{y} = f(\boldsymbol{x})$ on a dataset $\boldsymbol{x}$.
  3. Fit a simpler surrogate model g (e.g., linear regression, decision tree) to approximate f using $(\boldsymbol{x}, \hat{y})$.
  4. Evaluate and interpret g to understand the behavior of f.

- Global surrogate models are particularly useful when the original model is a "black box" (e.g., ensemble models, neural networks).

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Global Surrogate Models: Math

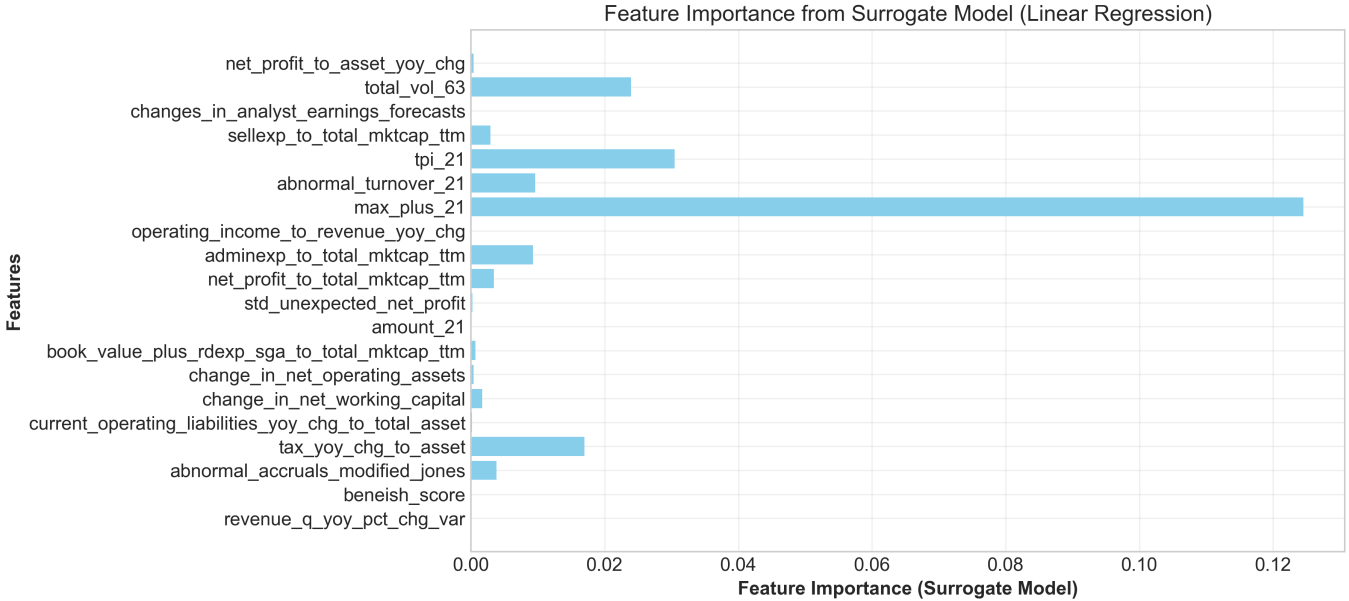
- Let $f(\boldsymbol{x})$ be the original complex model and $g(\boldsymbol{x})$ be the surrogate model.
- The surrogate model g is trained to minimize the discrepancy between f and g:

$$\min_{g} \frac{1}{n} \sum_{i=1}^{n} \left(f(\boldsymbol{x}_i) - g(\boldsymbol{x}_i) \right)^2 .$$

- Once trained, g serves as an interpretable approximation of f, and its structure can be analyzed to understand feature importance.

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Global Surrogate Models: Example



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LIME: Intuition

- Local interpretable model-agnostic explanations (LIME) is a local, model-agnostic explanation method designed to interpret **individual predictions**.
- LIME approximates the behavior of a complex model f in the vicinity of a specific instance $\boldsymbol{x}$ by fitting a simpler, interpretable model g (e.g., linear regression).
- Steps:
  1. Select an instance $\boldsymbol{x}$ to explain.
  2. Generate a neighborhood of perturbed samples $\boldsymbol{x}'$ around $\boldsymbol{x}$.
  3. Use the complex model f to predict $\hat{y}' = f(\boldsymbol{x}')$ for the perturbed samples.
  4. Fit a simpler surrogate model g (e.g., linear regression) to approximate f on the perturbed data $(\boldsymbol{x}', \hat{y}')$.
  5. Analyze g to understand how features influence f's prediction for $\boldsymbol{x}$.

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LIME: Math

- Let $f(\boldsymbol{x})$ be the original complex model and $g(\boldsymbol{x}')$ be the simpler surrogate model.
- LIME minimizes a weighted loss function to ensure g locally approximates f near $\boldsymbol{x}$:

$$\min_{g} \sum_{\boldsymbol{x}_i' \in \mathcal{N}(\boldsymbol{x})} \pi_{\boldsymbol{x}}(\boldsymbol{x}_i') \left(f(\boldsymbol{x}_i') - g(\boldsymbol{x}_i')\right)^2 + \Omega(g),$$

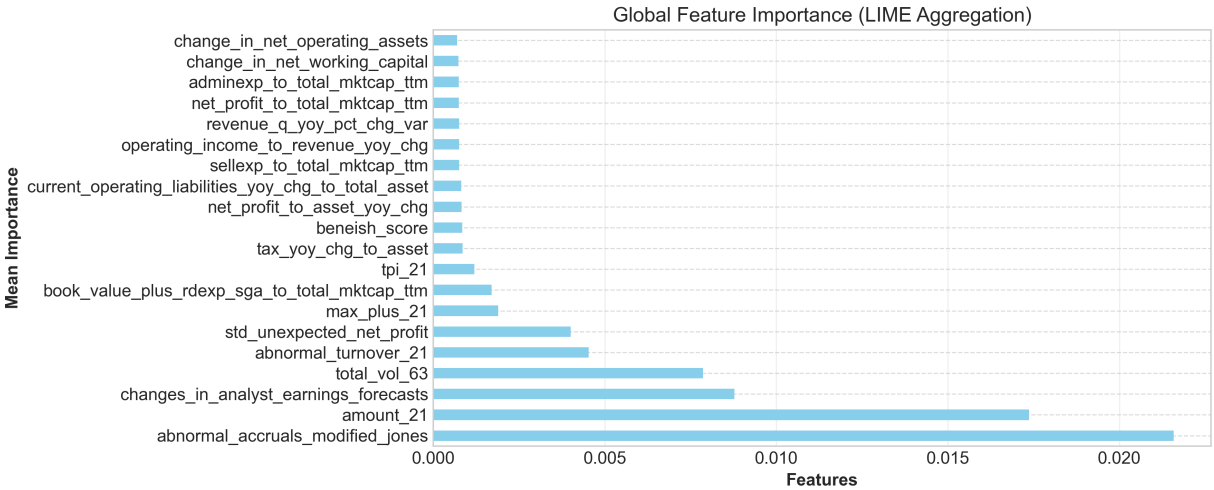
  where:

  — $\pi_{\boldsymbol{x}}(\boldsymbol{x}_i')$: a proximity weight function measuring the similarity between $\boldsymbol{x}$ and $\boldsymbol{x}_i'$.

  — $\Omega(g)$: a regularization term to promote interpretability of g (e.g., sparsity).

- The surrogate model g is then analyzed to explain the contribution of features to $f(\boldsymbol{x})$.

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- **Note: LIME is supposed to be used for interpreting individual predictions. The "aggregation" for LIME is purely for illustration purpose.**

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Shapley Values: Intuition

- Shapley Values originate from cooperative game theory and are used to fairly distribute the "payout" (e.g., model prediction) among the features based on their contributions.
- In the context of machine learning, Shapley Values quantify the contribution of each feature to **a specific prediction** by considering all possible subsets of features.
- Steps:
  1. Treat the prediction task as a cooperative game where the features are "players."
  2. For a given instance $\boldsymbol{x}$, compute the marginal contribution of each feature by including or excluding it from all possible subsets of features.
  3. Aggregate the marginal contributions across all subsets to calculate the Shapley Value for each feature.

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Shapley Values: Math

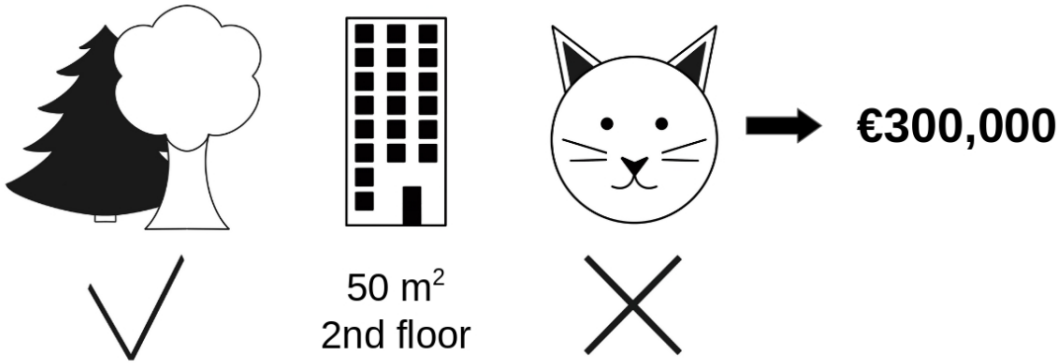
- Let $v(S)$ be the value function representing the contribution of a subset of features S to the model's performance or prediction.
- Let $\{1, 2, \ldots, M\} \setminus \{k\}$ represent the set of features **excluding** feature k.
- The Shapley Value ϕ_k for feature k is computed as:

$$\phi_k = \sum_{S \subseteq \{1,2,\ldots,M\} \setminus \{k\}} \frac{|S|!(M - |S| - 1)!}{M!} \Big[v(S \cup \{k\}) - v(S) \Big],$$

  where $|S|!(M - |S| - 1)!/M!$ is a weighting factor, ensuring all subsets are treated fairly. For features not in $S \cup \{k\}$, their values are randomly drawn from the dataset to maintain consistency with the data distribution.

- **To obtain global importance, we calculate the average of the absolute Shapley values for each feature across the entire dataset.**

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- Four features: `park`, `cat`, `area` and `floor`.
- This specific instance: `park:nearby`, `cat:banned`, `area:50` and `floor:2nd`.

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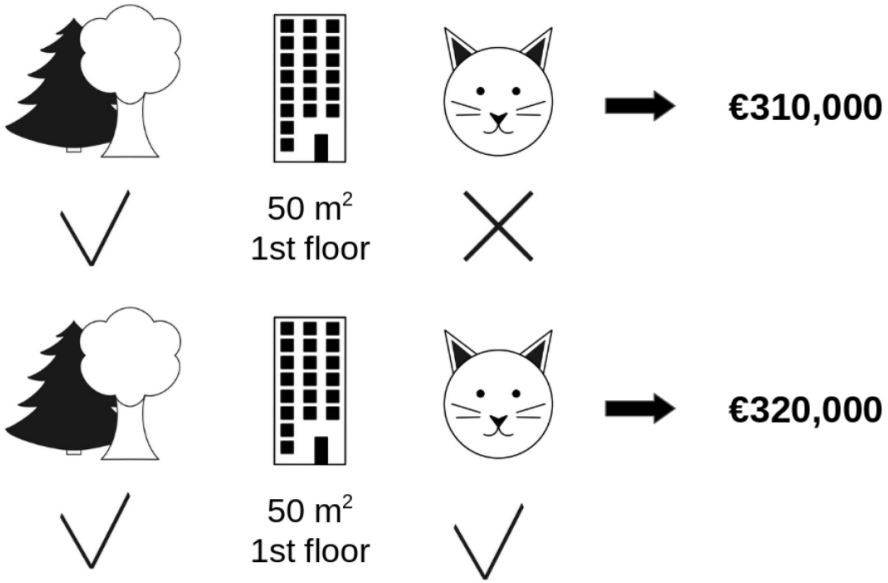


Example: Contribution of `cat:banned`

- Consider the coalition of `park` and `area`. Recall, for our instance, we have `nearby` and `50`.
- For $v(S \cup \{k\})$ and $v(S)$, replace `floor` for this instance with a randomly drawn value, e.g., `floor:1st`.
- For $v(S)$, replace `cat` with a randomly drawn value, e.g., `cat:allowed`.
- Predict apartment price:
  - $v(S \cup \{k\})$, i.e., `part:nearby`, `area:50`, `cat-banned`, and `floor:1st`: 310,000
  - $v(S)$, i.e., `part:nearby`, `area:50`, `cat-allowed` and `floor:1st`: 320,000
  - Marginal contribution of `cat-banned`: $310,000 - 320,000 = -10,000$.
- Repeat this process multiple times with different random values for better estimates.

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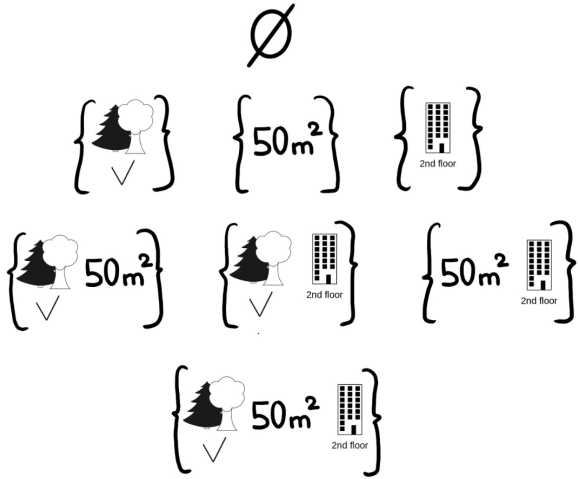
Figure: One Sample Repetition



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All Possible Coalitions

- To calculate the Shapley value for `cat-banned`, consider all possible coalitions:
  1. No feature values
  2. `park-nearby`
  3. `area-50`
  4. `floor-2nd`
  5. `park-nearby + area-50`
  6. `park-nearby + floor-2nd`
  7. `area-50 + floor-2nd`
  8. `park-nearby + area-50 + floor-2nd`
- For each coalition, calculate marginal contribution.
- Shapley value = Weighted average of marginal contributions.



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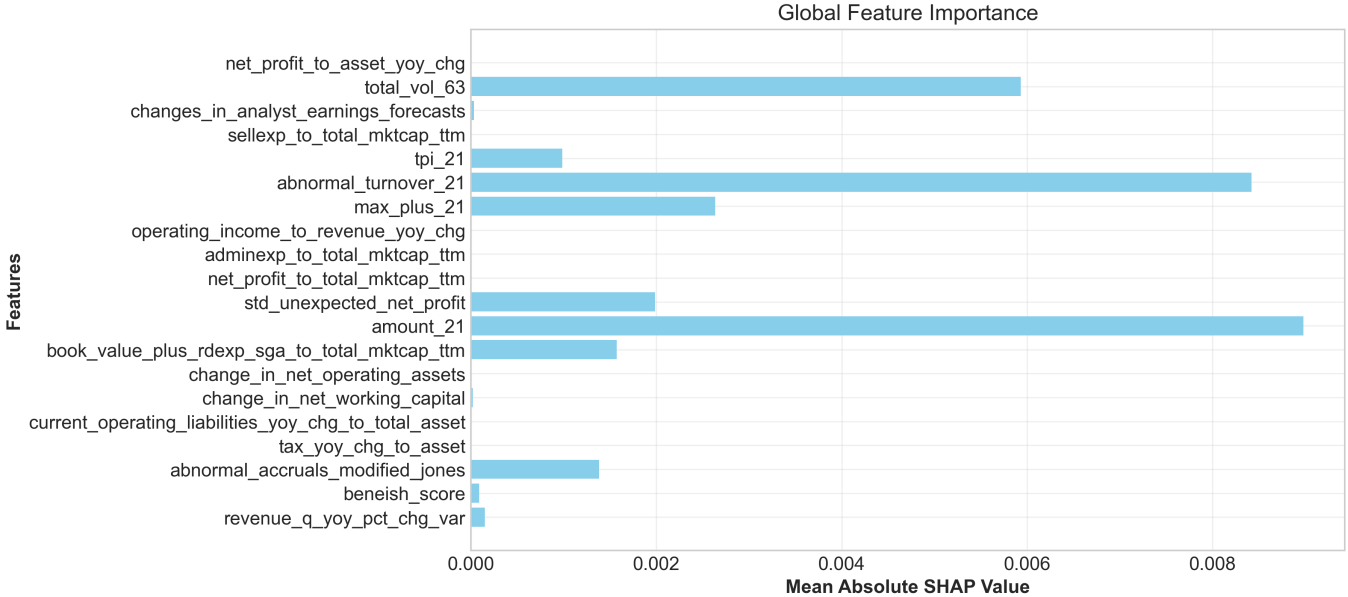


Computation Complexity

- Shapley value computation time increases exponentially with the number of features.
- Use sampling to compute contributions for only a subset of coalitions.
- This reduces computation time while maintaining reasonable accuracy.
- Recent advances makes calculating approximate Sharpley value feasible for complex models with many features, e.g., SHAP (SHapley Additive exPlanations) by Lundberg and Lee (2017).

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SHAP Example



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Feature Interaction: Intuition

- A method to assess feature importance by measuring how much the interaction between features contributes to the model's predictions.
- Steps:
  1. Select a pair (or group) of features $\{x_1, x_2, \dots\}$ to evaluate their interaction.
  2. Compute the model's predictions $f(\boldsymbol{x})$ with and without considering the interaction between these features.
  3. Quantify the interaction effect by comparing the difference in predictions.
  4. Aggregate interaction effects across the dataset to understand global feature interactions.
- **Feature interaction captures the dependency or synergy between two or more features in influencing the model's output.**

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Feature Interaction: Math

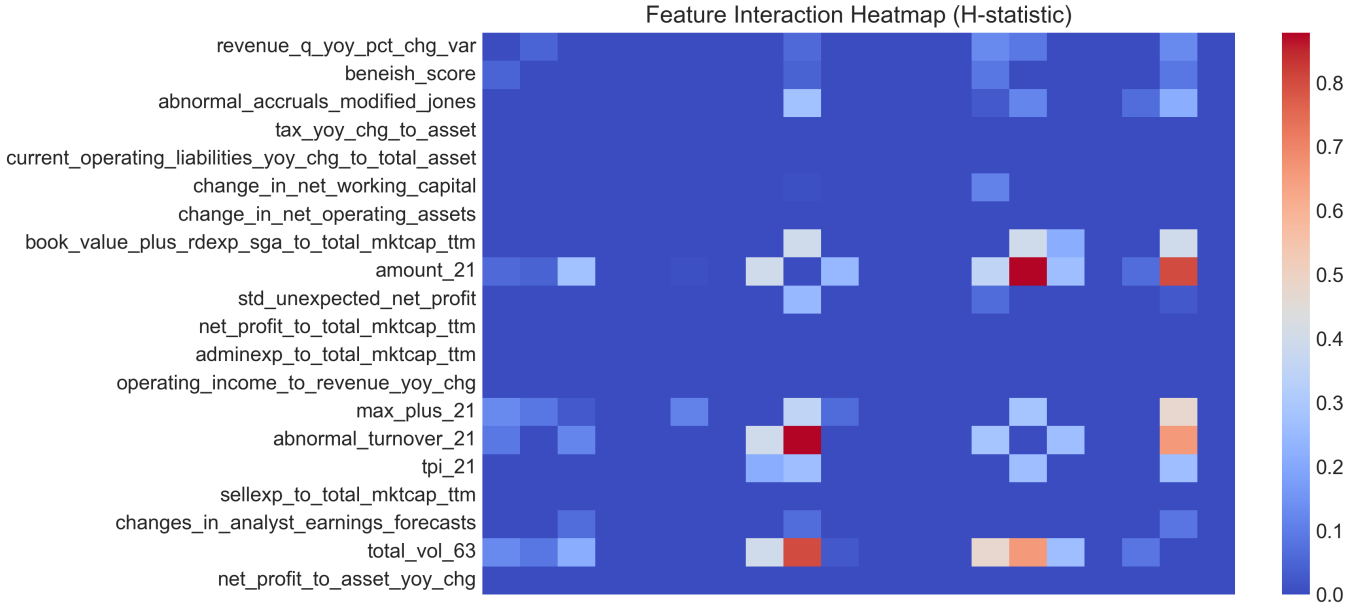
- Let $f(\boldsymbol{x})$ be the model and $\boldsymbol{x} = \{x_1, x_2, \ldots, x_K\}$ be the feature set.
- The interaction effect between features x_i and x_j for a specific instance can be quantified as:

$$I_{ij} = \mathbb{E}_{\boldsymbol{x}_{-ij}} \left[f(x_i, x_j, \boldsymbol{x}_{-ij}) - f(x_i, \boldsymbol{x}_{-ij}) - f(x_j, \boldsymbol{x}_{-ij}) + f(\boldsymbol{x}_{-ij}) \right],$$

  where $\boldsymbol{x}_{-ij}$ represents all features except x_i and x_j.
- This measures the contribution of the interaction between x_i and x_j to the model's predictions, isolating it from individual feature effects.
- Aggregating I_{ij} across the dataset provides a global view of feature interactions.

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Feature Interaction: Example



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Why Does the Choice Matter?

- Different methods, different assumptions:
  - Model-specific methods (e.g., coefficients, impurity reduction) rely on the **internal structure** of the model.
  - Model-agnostic methods (e.g., SHAP, permutation importance) work across all models but may require more computation.
- Strengths and limitations:
  - Model-specific methods are fast and interpretable but limited to certain model types.
  - Model-agnostic methods are flexible but computationally expensive and sometimes harder to interpret.

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How to Choose a Method?

4 Practical Considerations

- Consider model complexity:
  - Simple models (e.g., linear regression, decision trees): Use model-specific methods.
  - Black-box models (e.g., neural networks, ensemble models): Use model-agnostic methods.
- Define your interpretation goal:
  - For global understanding: Use global methods (e.g., permutation importance, PDP).
  - For individual predictions: Use local methods (e.g., LIME, SHAP).
- Evaluate computational cost:
  - Permutation importance and SHAP can be computationally expensive for large datasets.
  - Use simpler methods (e.g., impurity reduction) for quick assessments.

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Resolving Inconsistencies

- Understand the methods: Each method has its own assumptions and limitations.
- Look for consistent patterns:
  - Compare results across methods to identify robustly important features.
  - Investigate features with inconsistent importance scores.
- Check feature interactions and correlations:
  - High correlations between features can cause some methods to misattribute importance.
  - Use methods like SHAP or feature interaction analysis to account for these effects.

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Recommendations

4 Practical Considerations

- Start simple: Use simpler methods (e.g., permutation importance) for a quick overview of feature importance.
- Go deeper when needed: For black-box models or detailed analysis, use advanced methods.
- Validate with domain knowledge: Ensure feature importance results align with domain expertise and business context.
- Iterate: Reassess feature importance after refining the model or adding new data.

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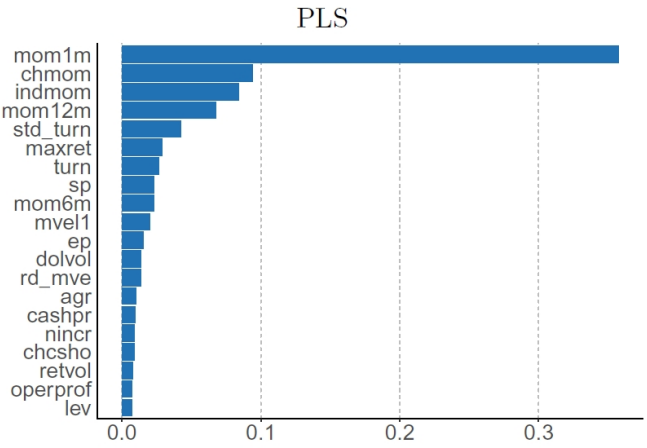


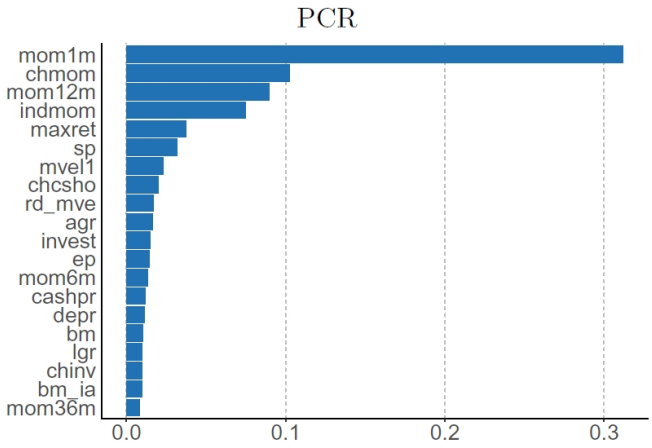
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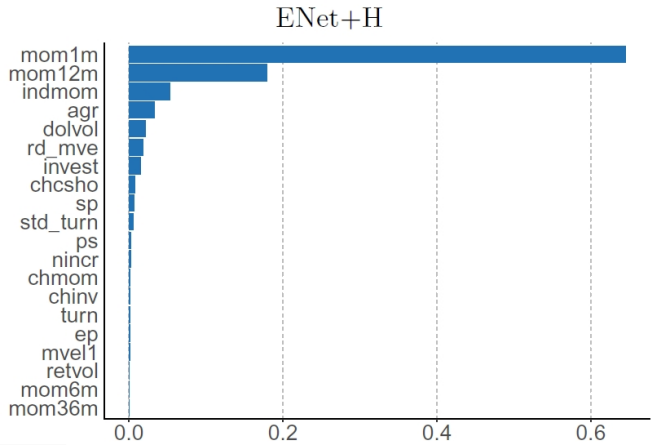
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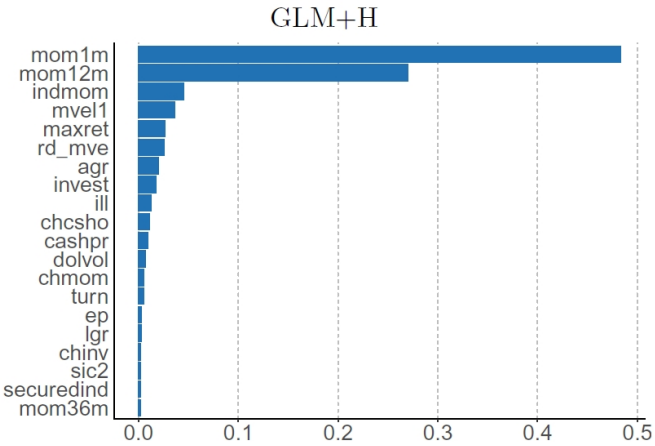
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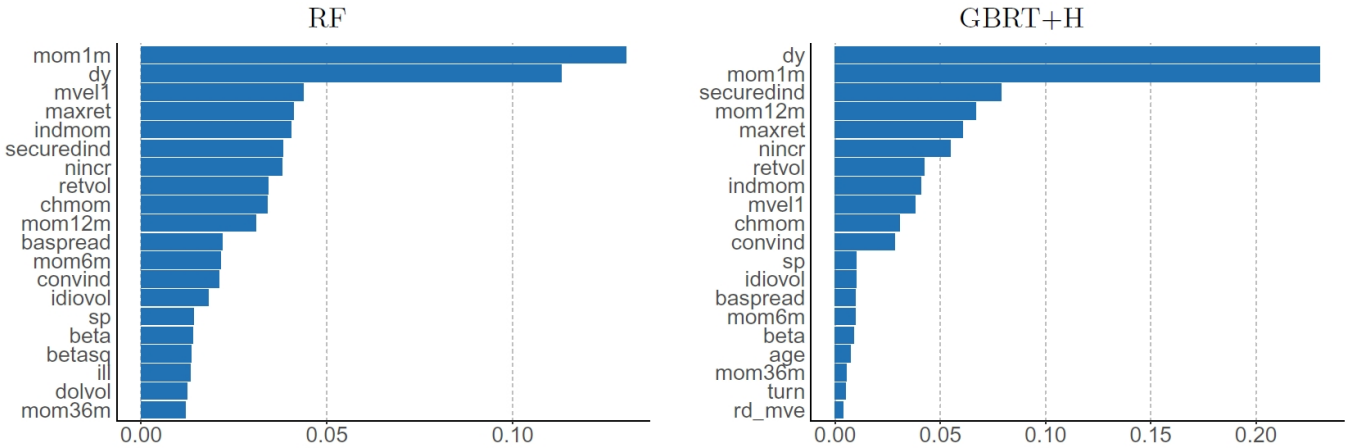




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GKX2020, Zero Permutation

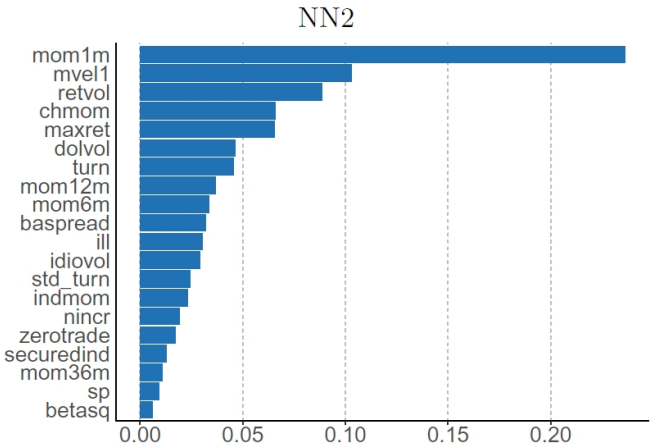
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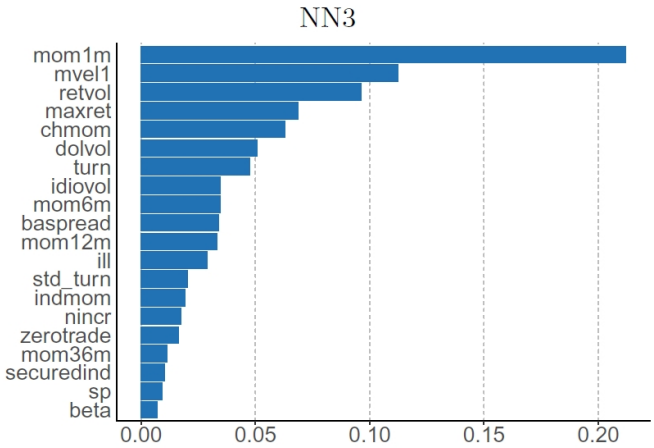


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GKX2020, Zero Permutation

5 Asset Pricing Applications

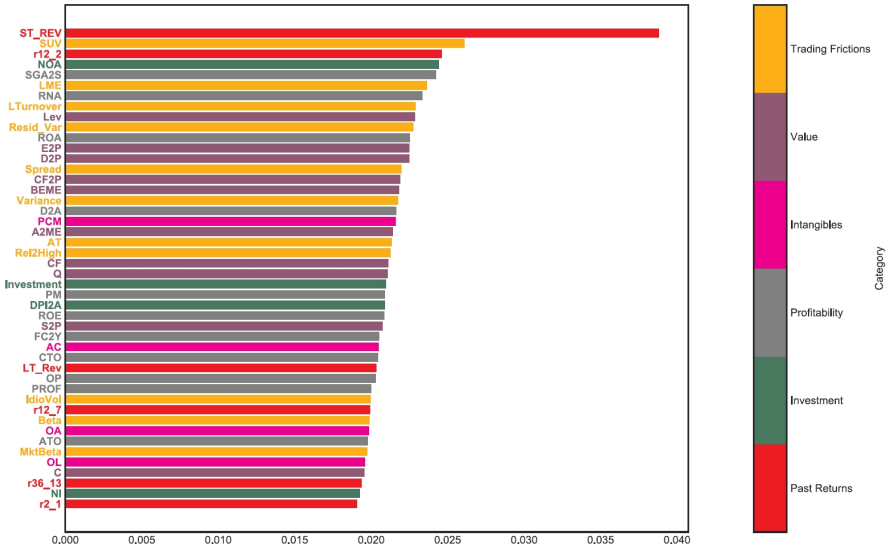




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CPZ2024, SDF's Sensitivity to Features

5 Asset Pricing Applications



Notes. The figure shows the GAN variable importance ranking of the 46 firm-specific characteristics in terms of average absolute gradient (VI) on the test data. The values are normalized to sum up to one.

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The Problem

- Missing covariates are an unavoidable challenge in quantitative investing, especially when dealing with alternative data.
- Missing data refers to the absence of specific covariate values for certain stocks at certain points in time.
- Popular methods for handling missing data:
  - Filling with cross-sectional mean/median.
  - Filling with industry mean/median.
  - Using the previous period's value (if available).
- Another crude approach is to exclude assets with missing data (sample selection bias).

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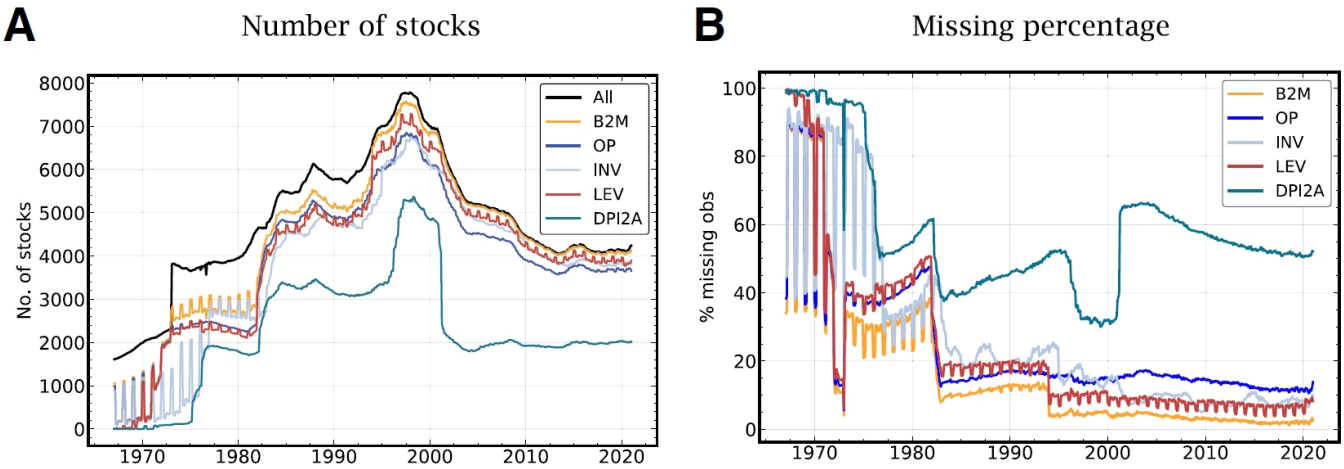


Key Questions to Address

- Is the missingness of data random across time and cross-section, or does it exhibit patterns (e.g., small-cap stocks missing more variables)?
- Do covariates across firms exhibit cross-sectional and time-series correlations that can be leveraged for better imputation?
- What are the implications of excluding missing data on:
  - Factor premium estimation?
  - Real-world investment opportunities (e.g., Sharpe ratio)?
- Two representative top journal papers are:
  - Bryzgalov et al. (2025) – Our focus.
  - Freyberger et al. (2025)

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Stylized Fact 1 – Missing Data Is Prevalent

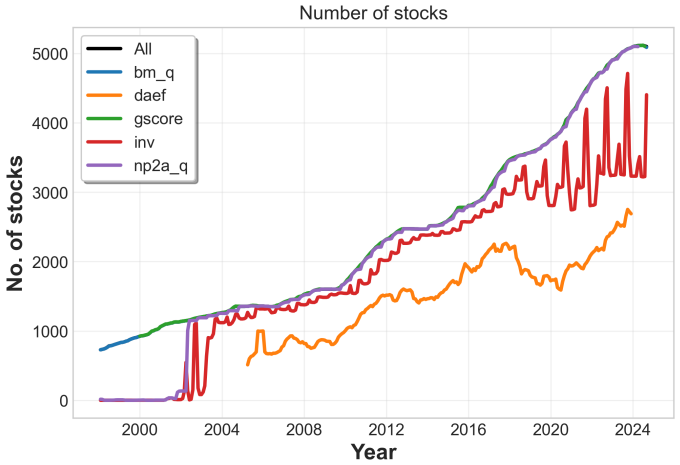


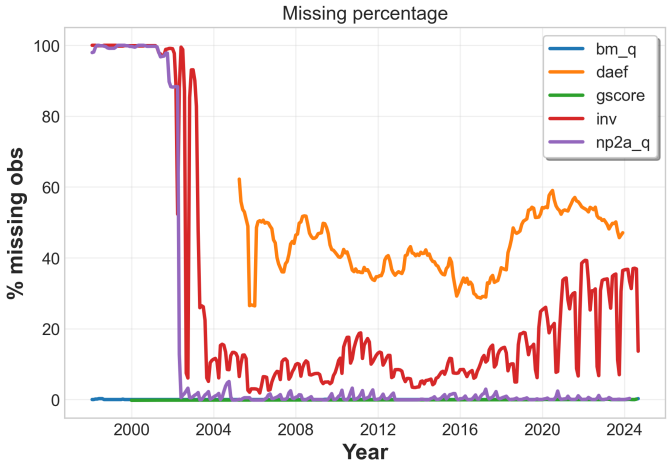
- Source: Bryzgalov et al. (2025)

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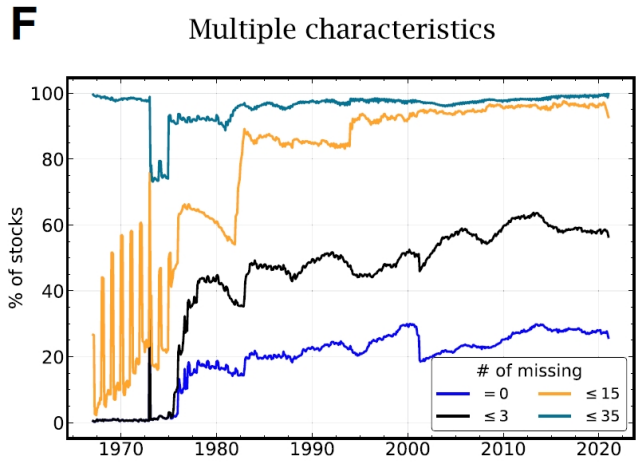
Stylized Fact 1 – Chinese Evidence

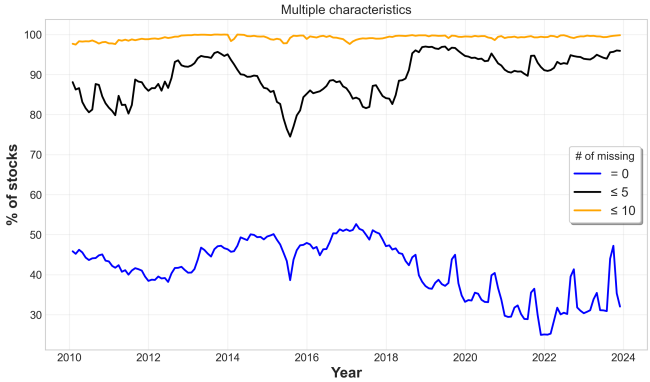
6 Missing Data





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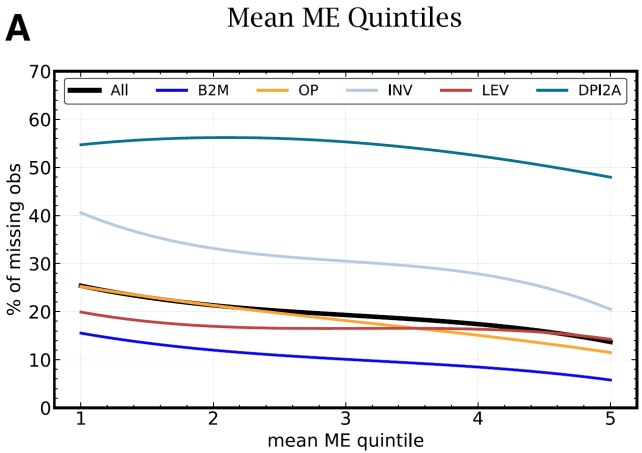


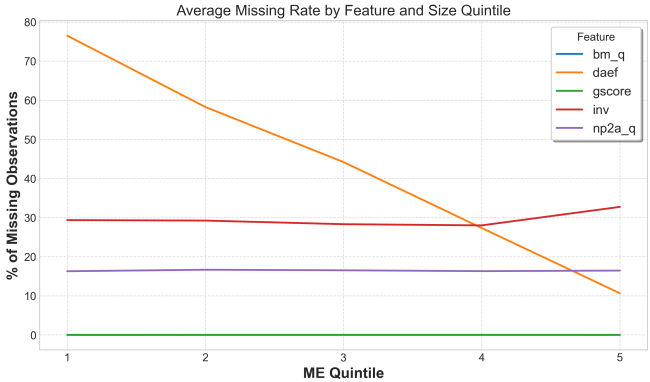


- Source: Bryzgalov et al. (2025) for US results

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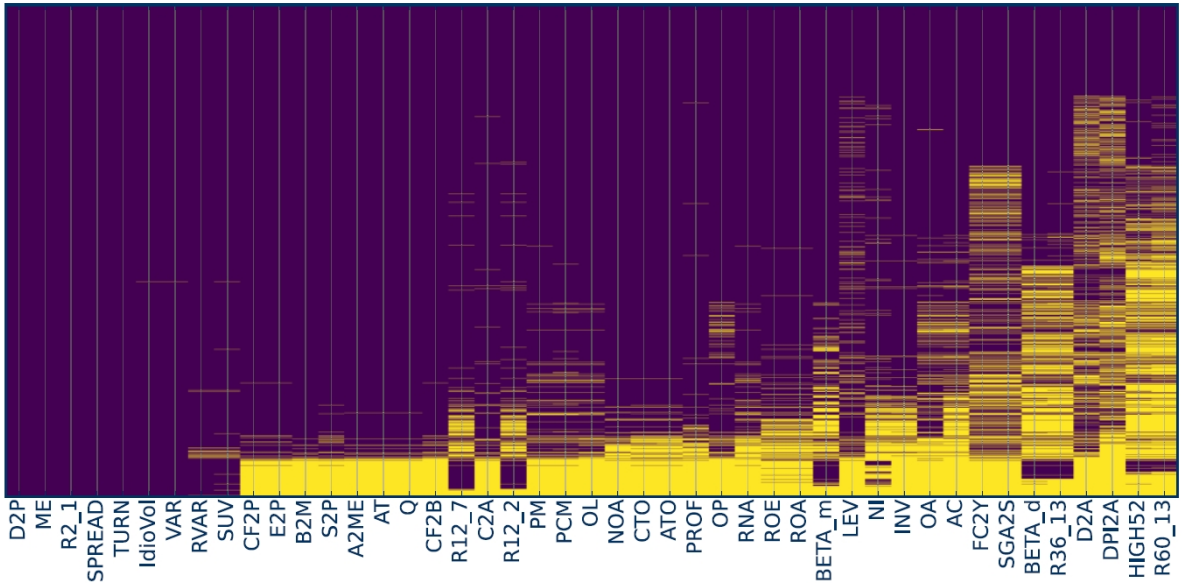
- Unbalanced missing pattern w.r.t. size (Bryzgalov et al. 2025 for US results):



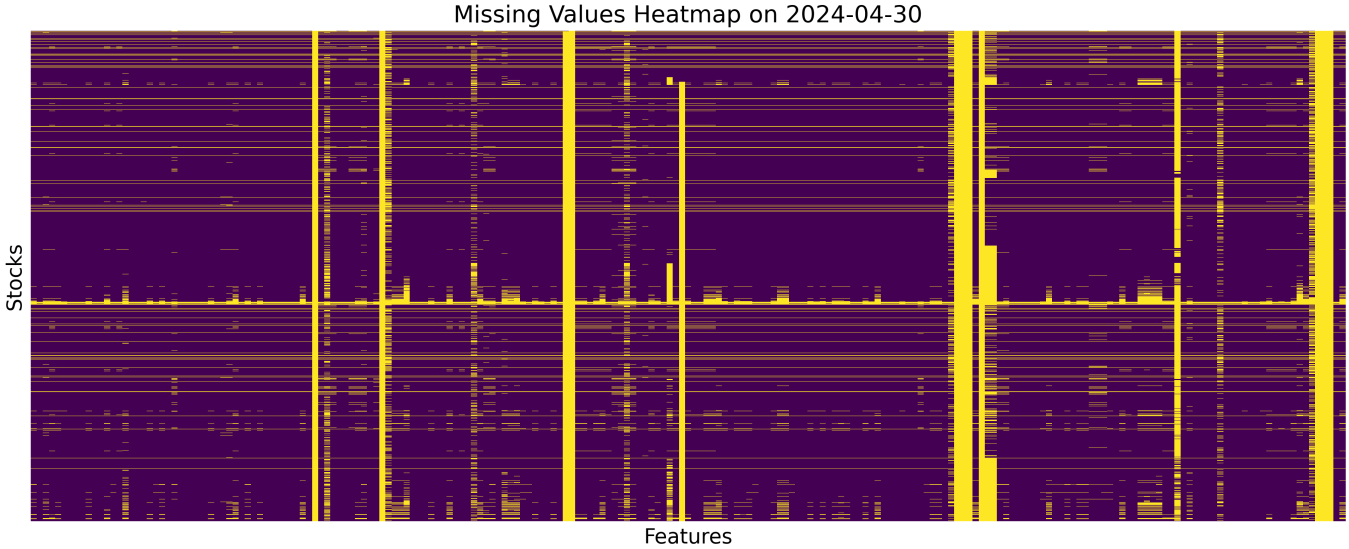


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Stylized Fact 3 – Not Completely Random



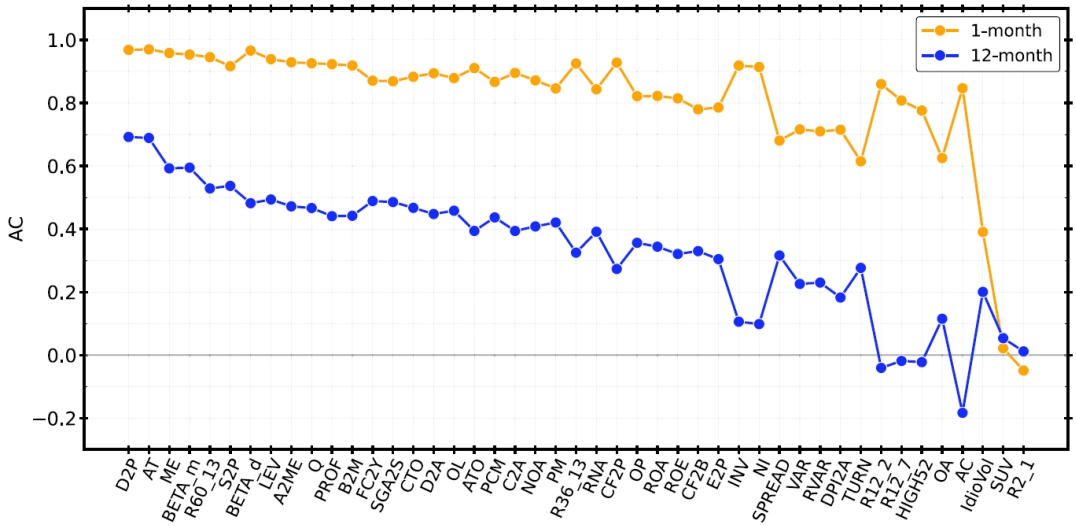
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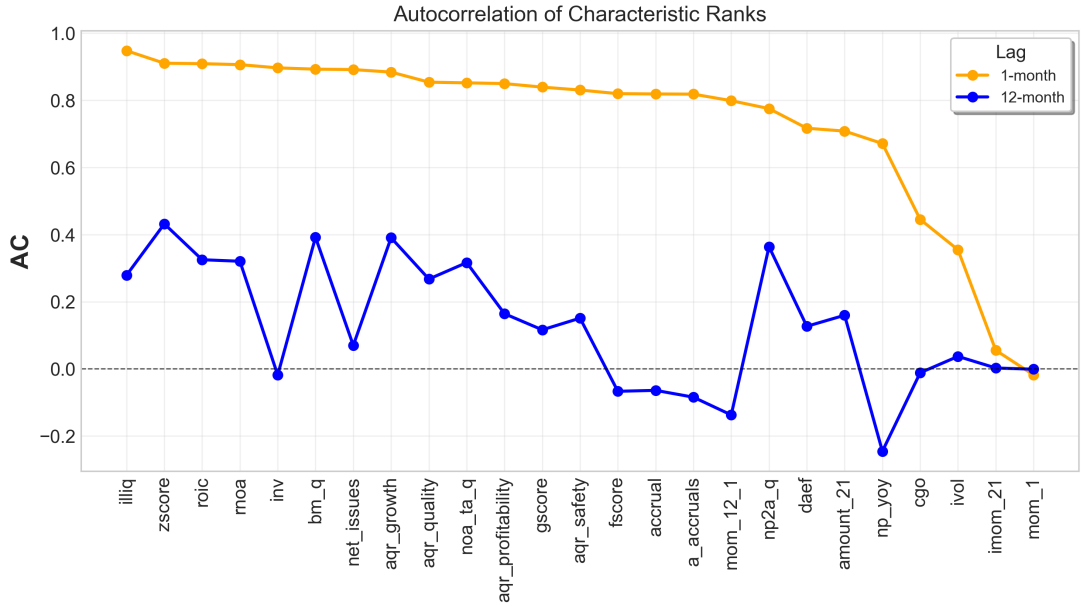
Stylized Fact 3 – Not Completely Random

- Time series autocorrelation of characteristic ranks (Bryzgalov et al. 2025):



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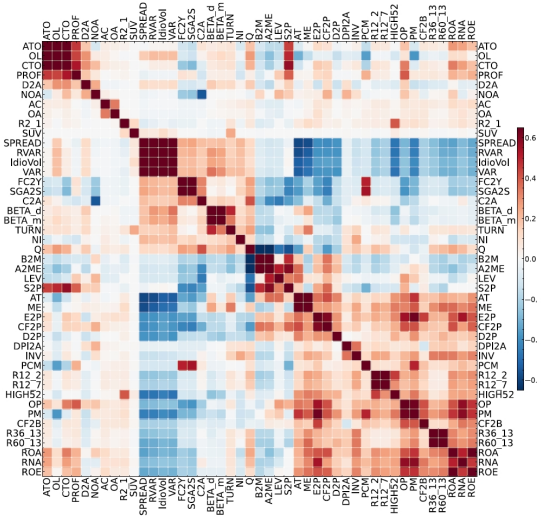
Stylized Fact 3 – Chinese Evidence

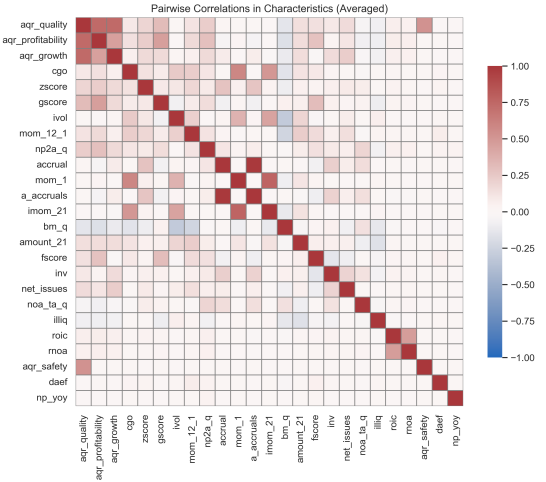


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Stylized Fact 3 – Not Completely Random

- Cross-sectional correlations (Bryzgalov et al. 2025 for US results):





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- Goal: Estimating a low-dimensional model to infer a characteristic value with observed cross-sectional information.

- Let $C_{i,l}^t$ represent the firm characteristic l for stock i at time t, with dimensions:
  - Cross-sectional stock dimension $i = 1, \cdots, N_t$,
  - Time-series dimension $t = 1, \ldots, T$,
  - Different characteristics $l = 1, \cdots, L$.

- The Model:

$$C_{i,l}^t = (\mathbf{\Lambda}_l^t)^\top \boldsymbol{F}_i^t + e_{i,l}^t,$$

  where $\boldsymbol{F}_i^t$ is the transpose of ith element of $N_t \times K$ matrix $\mathbf{F}^t$ (with K being the factor number), and $\mathbf{\Lambda}_l^t$ is the transpose of the lth element of the $L \times K$ matrix $\mathbf{\Lambda}^t$. (Both $\boldsymbol{F}_i^t$ and $\mathbf{\Lambda}_l^t$ represent column vectors.)

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The Solution – No Missing Value Case

- If no missing values, this is just a regular PCA.
- Let $\boldsymbol{C}_i^t$ represent the $L \times 1$ vector of firm characteristics of stock i at t.
- Apply eigen-decomposition to the following $L \times L$ matrix:

$$\boldsymbol{\Sigma} = \frac{1}{N_t} \sum_{i=1}^{N_t} \boldsymbol{C}_i^t (\boldsymbol{C}_i^t)^\top.$$

  Hint: Do you still remember the different expressions for the covariance matrix? (See the PCA lecture!)

- But, we have missing values...

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- The (l, p)th element in $L \times L$ matrix $\mathbf{\Sigma}$ represents the cross-sectional correlation of firm characteristics l and p.
- When there are missing values, it can be estimated as:

$$\hat{\mathbf{\Sigma}}_{l,p}^{t} = \frac{1}{|\mathcal{Q}_{l,p}^{t}|} \sum_{i \in \mathcal{Q}_{l,p}^{t}} C_{i,l}^{t} C_{i,p}^{t},$$

  where $\mathcal{Q}_{l,p}^{t}$ is the set of all stocks that are observed for two characteristics l and p at time t; and $|\mathcal{Q}_{l,p}^{t}| \leq N_t$ is the number of suck stocks. (The (l, p) subscript in $\hat{\mathbf{\Sigma}}_{l,p}^{t}$ represents the (l, p)th element.)
- Apply eigen-decomposition to $\hat{\mathbf{\Sigma}}^{t} = \left[\hat{\mathbf{\Sigma}}_{l,p}^{t}\right]$.

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The Solution – Missing Value Case

- Matrix $\mathbf{\Lambda}^t$ is a **scaled** version of the first K eigenvectors (why scaled?):

$$\hat{\mathbf{\Lambda}}^t = \hat{\mathbf{V}}^t(\hat{\mathbf{D}}^t)^{1/2},$$

  where $\hat{\mathbf{D}}^t$ is a diagonal matrix of the largest K eigenvalues, and $\hat{\mathbf{V}}^t$ is the corresponding eigenvectors.

- Estimate characteristic factors F_i^t from a regularized ridge regression of firm characteristics on loadings $\hat{\mathbf{\Lambda}}^t$:

$$\hat{\boldsymbol{F}}_i^t = \left(\frac{1}{L}\sum_{l=1}^{L} W_{i,l}^t \hat{\mathbf{\Lambda}}_l^t (\hat{\mathbf{\Lambda}}_l^t)^\top + \gamma \mathbf{I}_K\right)^{-1} \left(\frac{1}{L}\sum_{i=1}^{l} W_{i,l}^t \hat{\mathbf{\Lambda}}_l^t C_{i,l}^t\right),$$

  where $\gamma \geq 0$ is the regularization parameter and $W_{i,l}^t = 1$ if stock i has characteristic l observed at t, and $W_{i,l}^t = 0$ otherwise.

- Impute missing values by $(\hat{\mathbf{\Lambda}}_l^t)^\top \hat{\boldsymbol{F}}_i^t$.

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- The procedure goes like this:
  1. Estimate XS model $(\hat{\boldsymbol{\Lambda}}_l^t)^\top \hat{\boldsymbol{F}}_i^t$ for each t.
  2. If $C_{i,l}^{t-1}$ is observed, estimate the residual $\hat{e}_{i,l}^{t-1} = C_{i,l}^{t-1} - (\hat{\boldsymbol{\Lambda}}_l^{t-1})^\top \hat{\boldsymbol{F}}_i^{t-1}$.
  3. Construct the explanatory covariates:

$$\boldsymbol{x}_{i,l}^t = \left[(\hat{\boldsymbol{\Lambda}}_l^t)^\top \hat{\boldsymbol{F}}_i^t, C_{i,l}^{t-1}, \hat{e}_{i,l}^{t-1}\right]^\top.$$

  4. Estimate $\boldsymbol{\beta}$ coefficient by cross-sectional regression:

$$\hat{\boldsymbol{\beta}} = \left(\sum_{i=1}^{N_t} W_{i,l}^t \boldsymbol{x}_{i,l}^t (\boldsymbol{x}_{i,l}^t)^\top\right)^{-1} \left(\sum_{i=1}^{N_t} W_{i,l}^t \boldsymbol{x}_{i,l}^t C_{i,l}^t\right),$$

     where $W_{i,l}^t = 1$ if both $\boldsymbol{x}_{i,l}^t$ and $C_{i,l}^t$ are observed and 0 otherwise.
  5. Impute missing values by $\hat{\boldsymbol{\beta}}^\top \boldsymbol{x}_{i,l}^t$.

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- Well, maybe not a few...

Table 2
Different imputation methods

Method	Estimation
Backward-Forward-XS (BF-XS)	$\hat{C}_{i,l}^{t,\text{BF-XS}} = (\hat{\beta}^{\text{BF-XS}})^\top \left(\hat{F}_i^{t,\gamma} (\hat{\Lambda}_l^t)^\top \quad C_{i,l}^{t-1} \quad \hat{e}_{i,l}^{t-1} \quad C_{i,l}^{t+1} \quad \hat{e}_{i,l}^{t+1} \right)$
Backward-XS (B-XS)	$\hat{C}_{i,l}^{t,\text{B-XS}} = (\hat{\beta}^{l,\text{B-XS}})^\top \left(\hat{F}_i^t (\hat{\Lambda}_l^t)^\top \quad C_{i,l}^{t-1} \quad \hat{e}_{i,l}^{t-1} \right)$
Forward-XS (F-XS)	$\hat{C}_{i,l}^{t,\text{F-XS}} = (\hat{\beta}^{l,\text{F-XS}})^\top \left(\hat{F}_i^t (\hat{\Lambda}_l^t)^\top \quad C_{i,l}^{t+1} \quad \hat{e}_{i,l}^{t+1} \right)$
Cross-sectional (XS)	$\hat{C}_{i,t}^{\text{XS}} = \hat{F}_i^{t,\gamma} (\hat{\Lambda}_l^t)^\top$
Time series (B)	$\hat{C}_{i,l}^{t,\text{B}} = \hat{\beta}^{l,\text{B}} C_{i,l}^{t-1}$
Previous value (PV)	$\hat{C}_{i,l}^{t,\text{PV}} = C_{i,l}^{t-1}$
Cross-sectional median	$\hat{C}_{i,l}^{t,\text{median}} = 0$

This table summarizes the different estimation approaches. Each estimation approach has a local and global version.

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- Out-of-sample RMSE is calculated by intentionally masking a portion of observed characteristics (randomly as OOS MAR or sequentially as OOS Block), modeling with the remaining data, and evaluating on the masked data.

Table 4: Imputation Error for Different Imputation Methods

	In-Sample			OOS MAR			OOS Block		
Method	all	quarterly	monthly	all	quarterly	monthly	all	quarterly	monthly
global BF-XS	**0.11**	**0.10**	**0.13**	**0.15**	**0.15**	**0.14**	**0.17**	**0.16**	**0.19**
global F-XS	0.10	0.07	0.14	0.16	0.17	0.16	0.18	0.17	0.20
global B-XS	0.15	0.15	0.14	0.16	0.16	0.15	0.19	0.18	0.20
global XS	0.19	0.18	0.21	0.23	0.22	0.24	0.22	0.21	0.24
global B	0.16	0.17	0.15	0.17	0.17	0.15	0.21	0.20	0.22
local B-XS	**0.15**	**0.16**	**0.14**	**0.16**	**0.17**	**0.15**	**0.19**	**0.19**	**0.20**
local XS	0.21	0.20	0.22	0.23	0.22	0.24	0.23	0.22	0.24
prev	0.18	0.18	0.18	0.19	0.19	0.19	0.23	0.21	0.25
local B	0.16	0.17	0.15	0.17	0.17	0.15	0.21	0.20	0.22
XS-median	0.29	0.29	0.29	0.29	0.29	0.29	0.28	0.28	0.29
ind-median	0.29	0.29	0.29	0.29	0.29	0.29	0.28	0.28	0.29

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Questions?

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